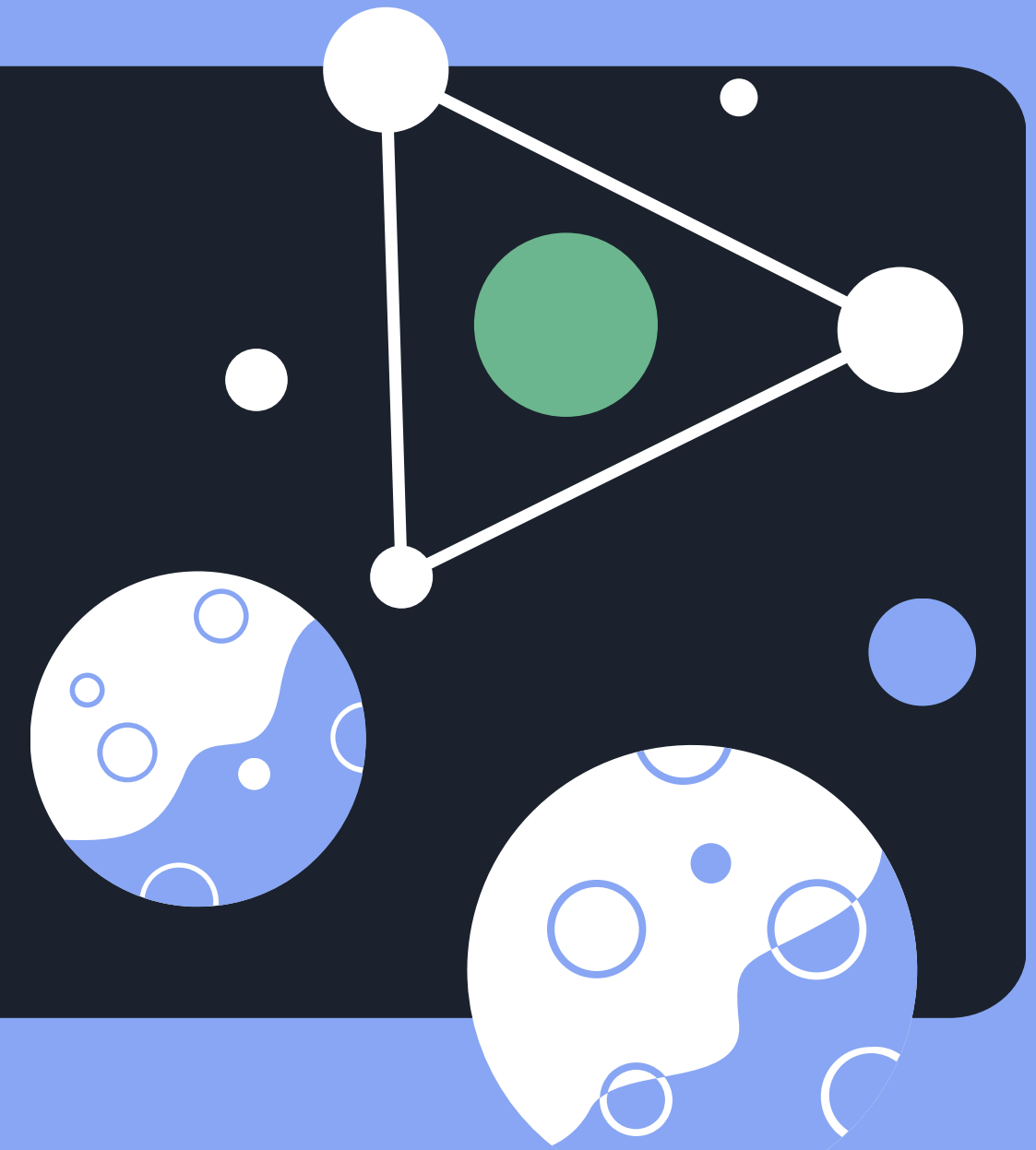




SecureFocus: A Secure Productivity and Collaboration Suite



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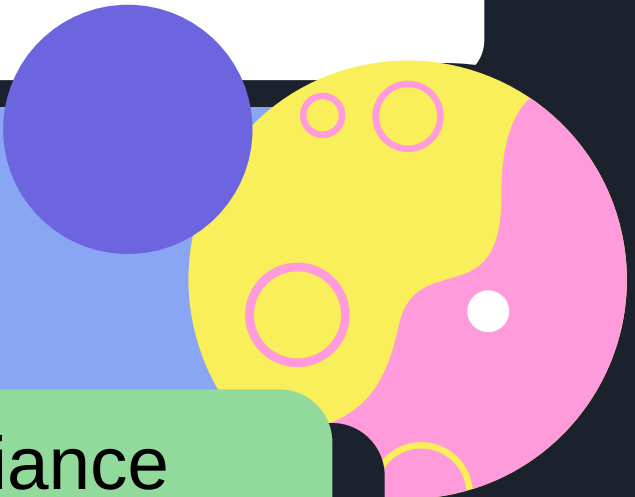
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PROBLEM STATEMENT

Many collaboration platforms prioritize usability over built-in security.

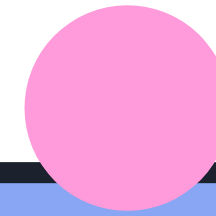
Remote work and cloud reliance increase risks of data leaks and unauthorized access.

Our challenge: How can teams collaborate efficiently while keeping sensitive data safe?

SecureFocus integrates encryption, MFA, and secure storage into one secure workspace



OUR OBJECTIVES



Private teamwork

Strong Authentication

Secure Messaging

Secure file sharing

**Data integrity &
Non-repudiation**

Availability

METHODOLOGY AND THE PROCESS MODEL

Incremental model

We used the Incremental model with Agile principles for flexibility and continuous improvement.

Each iteration added new features while refining security.

Why not others?

This model was chosen because it allows early feedback, fast adaptation, and parallel growth of functionality and security.



● ● ● METHODOLOGY AND THE PROCESS MODEL

Methodology

SQUARE (Security Quality Requirements Engineering):

Defines, elicits, and validates security requirements systematically.

Steps: Define → Identify Goals → Analyze Risks → Prioritize → Validate.

STRIDE Threat Modeling:

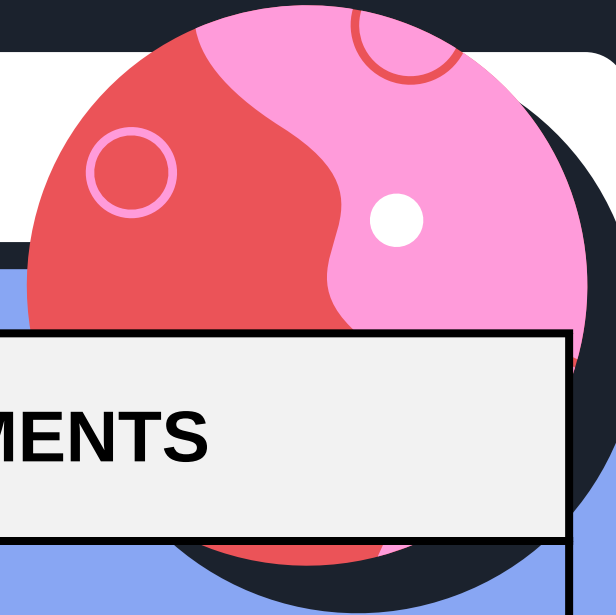
Identifies threats under: Spoofing, Tampering, Repudiation, Information Disclosure, DoS, Elevation of Privilege.

Helps map vulnerabilities to mitigations early.





REQUIREMENTS



FUNCTIONAL REQUIREMENTS	NON-FUNCTIONAL REQUIREMENTS
User registration & MFA-secured login	Performance
Project and task creation (Kanban view)	Usability: Intuitive interface
Encrypted file upload & sharing	Reliability: Backup and recovery mechanisms
Real-time chat with end-to-end encryption	Maintainability: Modular structure with clear documentation
Notifications & email alerts	Portability: Cross-platform and browser compatible
Role-based access (Admin, Member)	Security: HTTPS/TLS, MFA, encryption at rest
Audit logs & responsive web interface	Compliance: Follows GDPR principles



SECURITY REQUIRMENTS

**Strong passwords &
MFA for authentication**

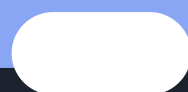
**Data encryption
(TLS & AES-256)**

**Role-Based Access
Control**

**Input validation
against XSS/SQLi**

**Audit logs & login alerts
for accountability**

**Regular backups &
brute-force protection**

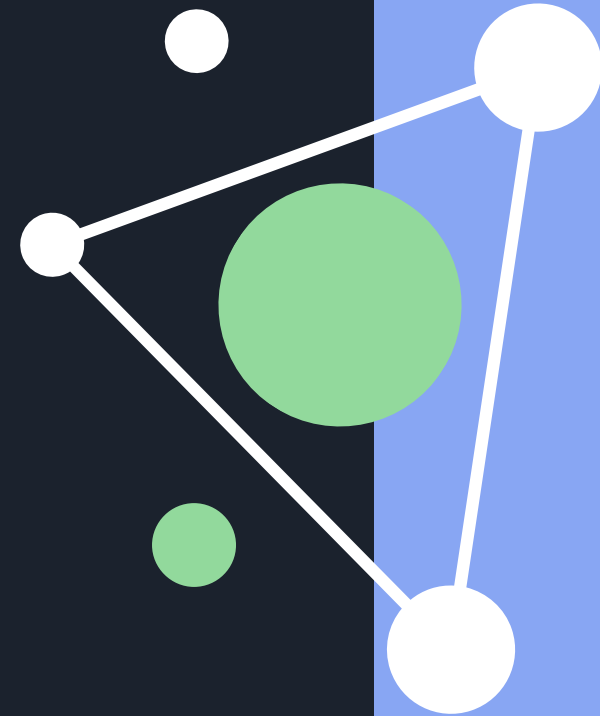




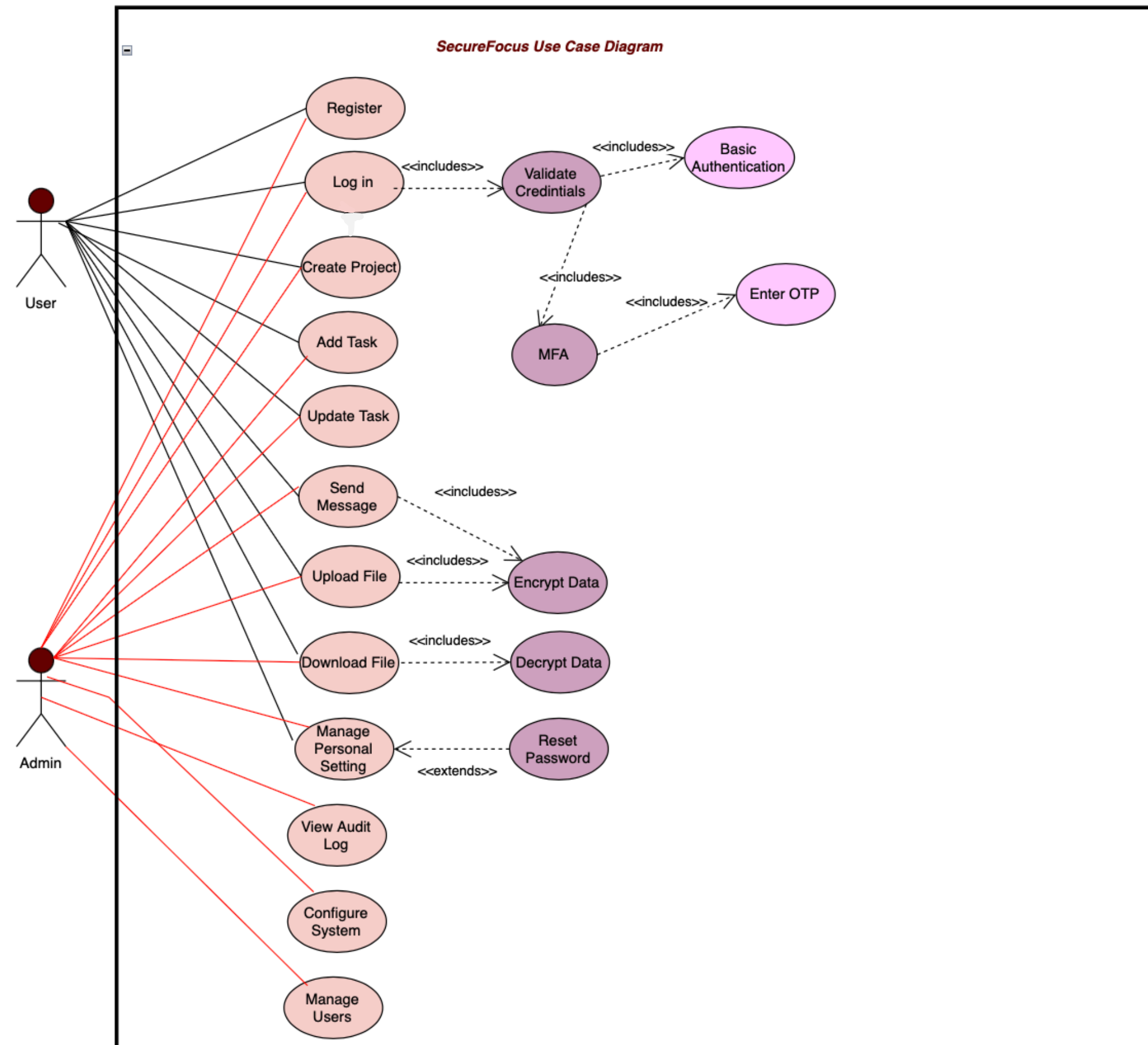
SYSTEM ARCHITECTURE

Three-Tier Design:

- **Frontend:** Browser-based user interface.
- **Backend:** Application logic, authentication, encryption, RBAC.
- **Database:** Stores encrypted messages, files, and audit logs.
- **Security Layers:** HTTPS, AES encryption, input validation, and server-side filters.



USE CASE DIAGRAM



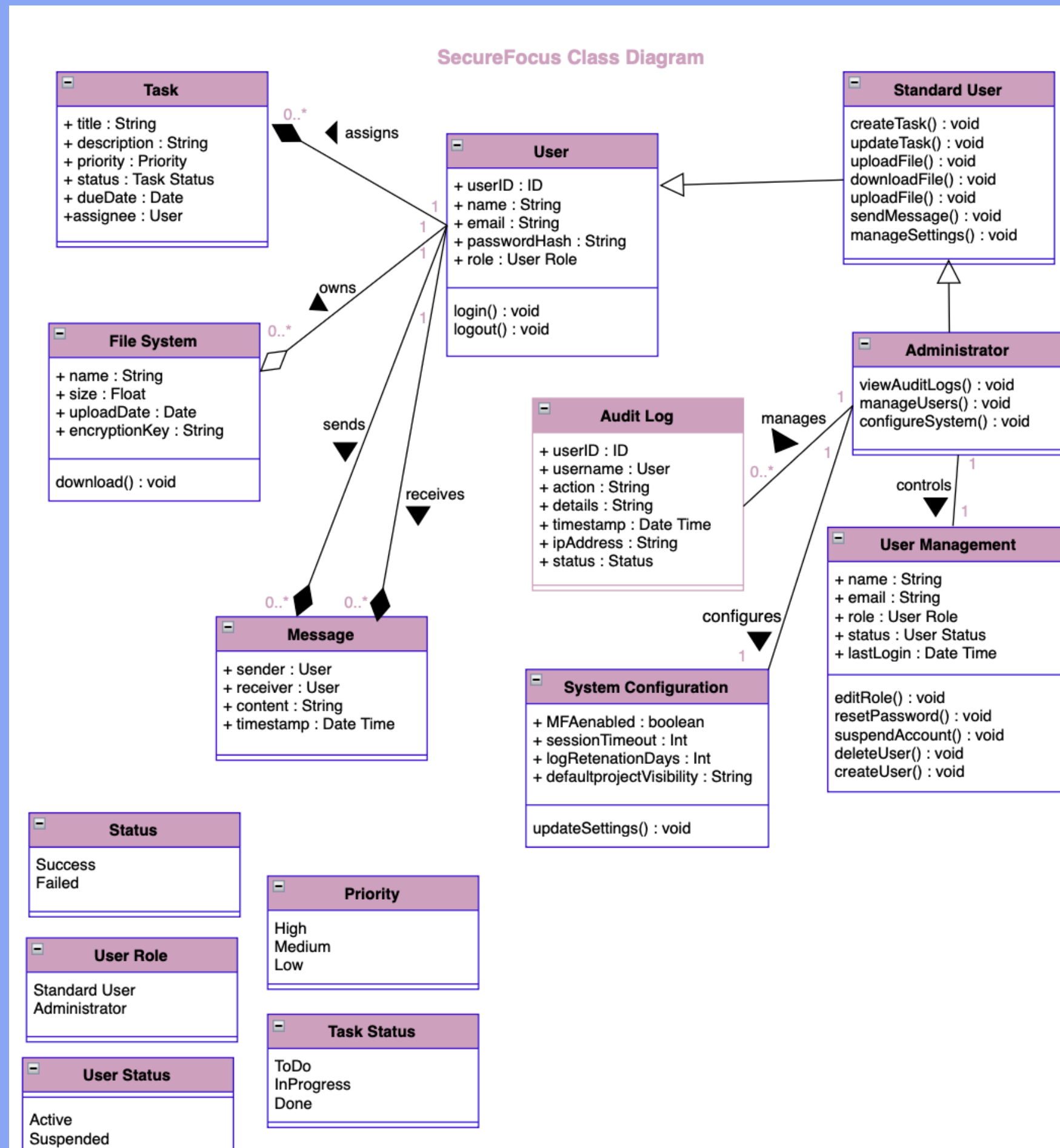
Two main actors:

User and Admin

- **Users can:**
register, log in, manage tasks, send messages, and upload files
- **Admins can :**
manage users and view audit logs



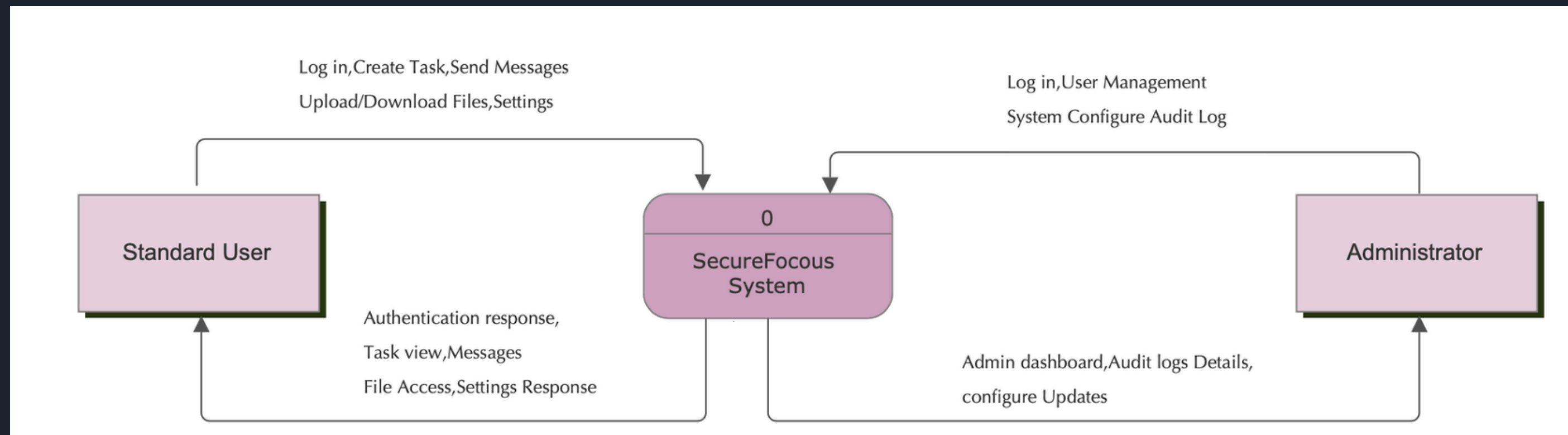
CLASS DIAGRAM



Represents the core
structure of the system

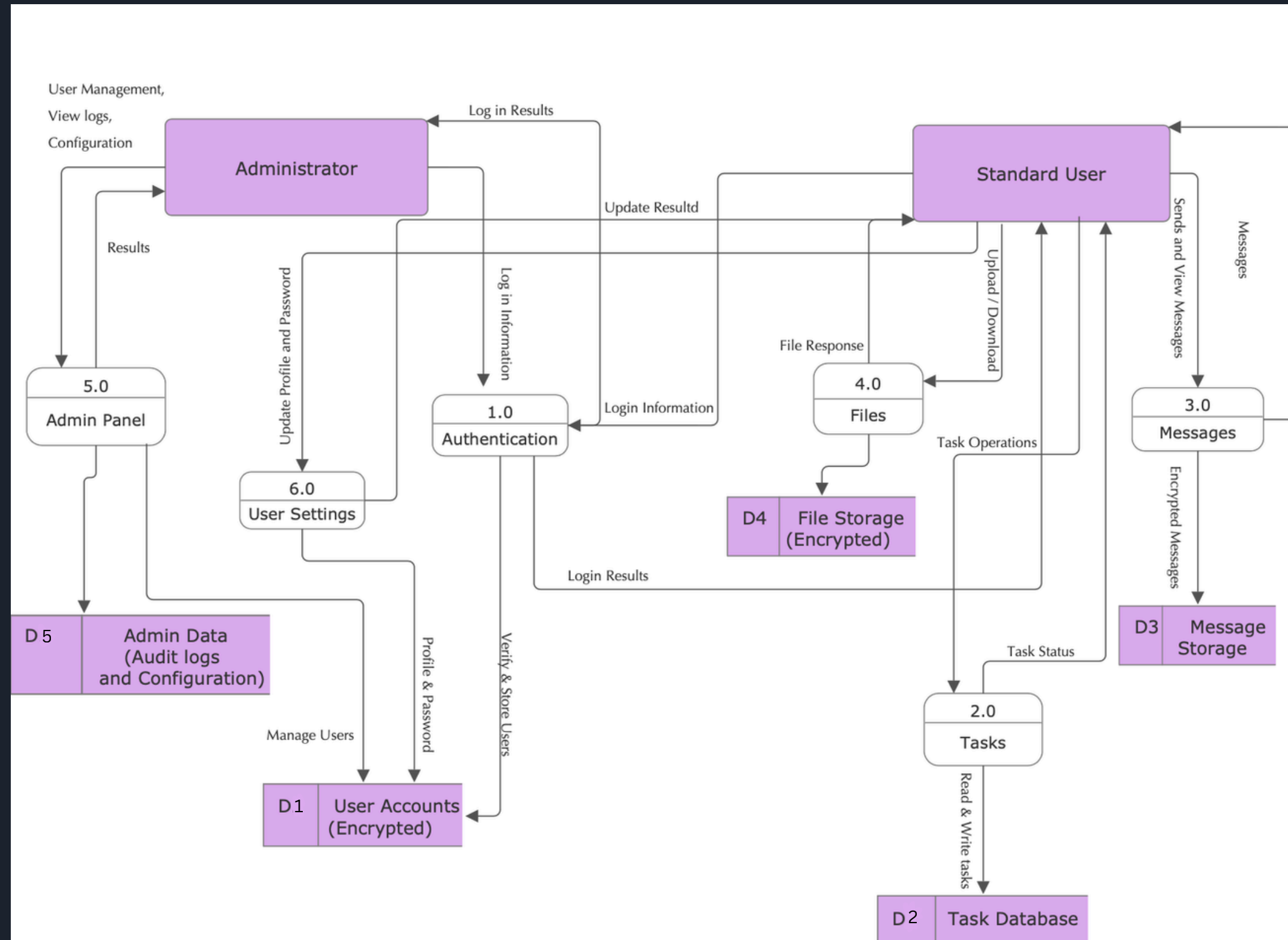


DATA FLOW DIAGRAM -LEVEL 0





DATA FLOW DIAGRAM -LEVEL 1





FIGMA-STANDARD USER VIEW



SecureFocus

A Secure Productivity and Collaboration Suite

Access your secure workspace

🔒 End-to-End Encrypted

Email Address

your.email@example.com

Password

Enter your password

Login As

👤 Standard User

🛡️ Administrator

Forgot Password?

🔒 Sign In Securely

Protected by:

🛡️ 256-bit AES • 🛡️ TLS 1.3 • 2FA Ready



SecureFocus

🏠 Dashboard

💬 Chat

📁 Files

🚪 Logout

Task Dashboard

Manage and track your team's projects

+ Add Task

Status



All Statuses

Team Member

All Members

View Mode

Kanban Board

To Do

2

Design new dashboard layout

Create mockups for the analytics dashboard

🟡 medium To Do

👤 Mike Johnson

📅 15/11/2025

Database optimization

Improve query performance for reports

🟢 low To Do

👤 Mike Johnson

📅 20/11/2025

In Progress

2

Update security protocols

Review and update authentication mechanisms

🔴 high In Progress

👤 Sarah Chen

📅 10/11/2025

Implement end-to-end encryption

Add E2EE for file sharing module

🔴 high In Progress

👤 Sarah Chen

📅 08/11/2025

Done

1

User testing session

Conduct usability testing with 5 participants

🟡 medium Done

👤 Emma Davis

📅 05/11/2025



+ Add Task

Team Member

All Members

View Mode

Kanban Board

Create New Task

Add a new task to your project board

Task Title

Enter task title

Description

Describe the task

Priority

Medium

Low

Medium

High

Assignee

Team member

Cancel

Create Task

User testing session

Conduct usability testing with 5 participants

medium

Done

Emma Davis

05/11/2025

The screenshot displays the SecureFocus Secure Chat application. At the top, the header includes the SecureFocus logo, navigation links for Dashboard, Chat, Files, and Logout, and a status indicator for E2E Encrypted messaging.

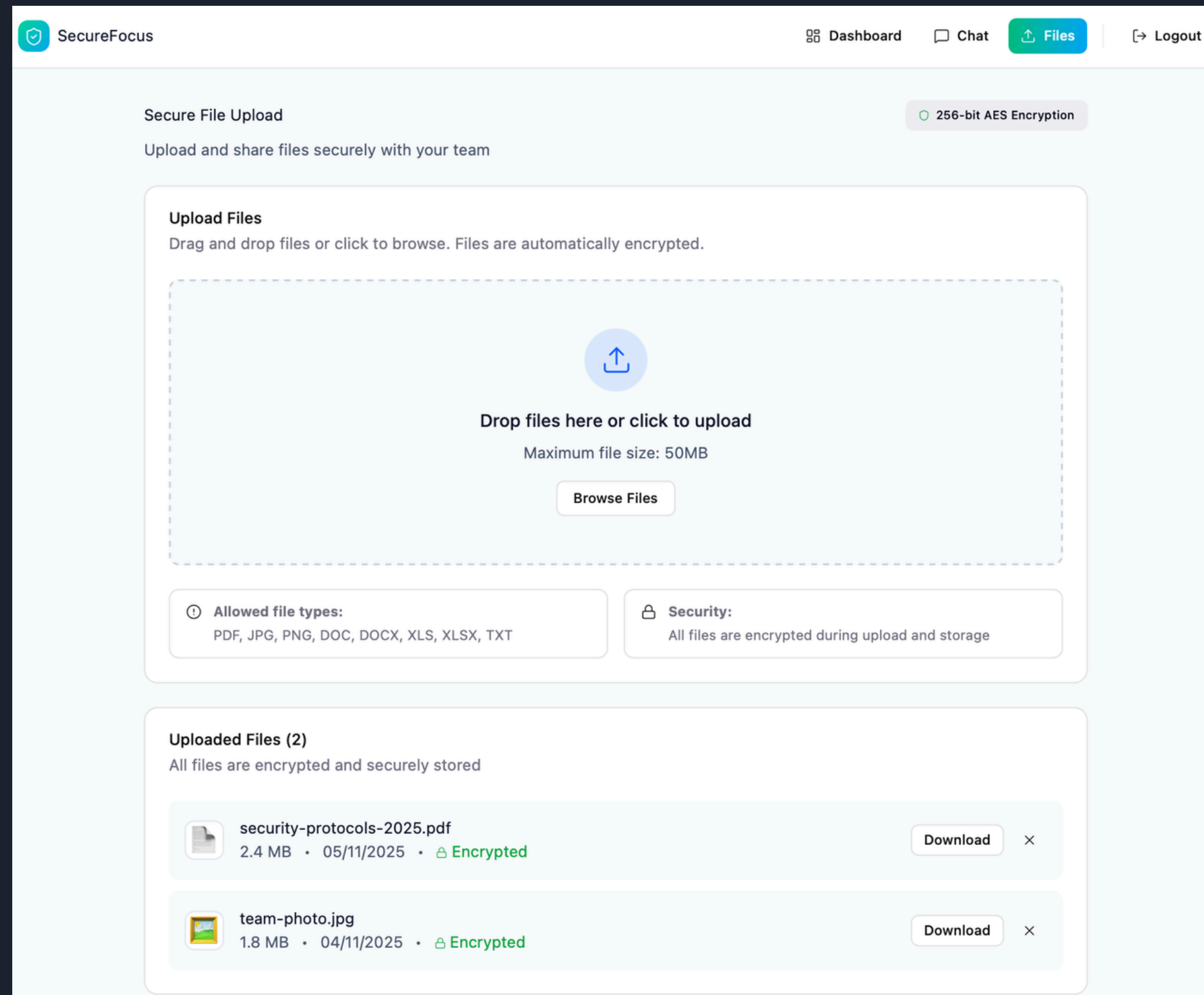
The main interface is divided into two sections. On the left is a sidebar titled "Conversations" with a search bar. It lists four active conversations: Sarah Chen (SC), Project Team (PT), Mike Johnson (MJ), and Security Team (ST). The Sarah Chen conversation is selected and highlighted.

The right section shows the chat window for Sarah Chen. It features a header with the contact name, a status icon, and a confirmation that "Messages are encrypted". The chat history shows a sequence of messages: Sarah Chen asks to review security protocols, the user responds affirmatively, Sarah Chen asks for implementation by Friday, and the user promises completion by Thursday afternoon. The chat concludes with Sarah Chen thanking the user for the update.

At the bottom, a footer message states: "Your messages are protected with end-to-end encryption". Below this is a text input field labeled "Type a secure message..." and a green "Send" button.



FIGMA-STANDARD USER VIEW





FIGMA-ADMINISTRATOR VIEW



SecureFocus

A Secure Productivity and Collaboration Suite

Access your secure workspace

End-to-End Encrypted

Email Address

your.email@example.com

Password

Enter your password

Login As

Standard User

Administrator

Forgot Password?

Sign In Securely

Protected by:

256-bit AES • TLS 1.3 • 2FA Ready

SecureFocus

Dashboard Chat Files Admin Logout

Admin Dashboard

Secure system management and monitoring

Admin Access

Dashboard Audit Logs User Management System Config

Total Users
5

Active Users
4

Failed Logins (24h)
1

Recent Actions
0

System Health Status

Current system status and security metrics

Authentication System
All services operational

Database Connection
Response time: 45ms

Encryption Services
256-bit AES encryption active

Quick Actions

Common administrative tasks

Add New User
Create user account

View Audit Logs
Review system activity

System Settings
Configure security



FIGMA-ADMINISTRATOR VIEW

Admin Dashboard

Secure system management and monitoring

Admin Access

[Dashboard](#) [Audit Logs](#) [User Management](#) [System Config](#)

Audit Log Panel

Comprehensive system activity tracking and monitoring

Sensitive Data

Search logs by user, action, or details...

All Actions

All Status

User ID	User Name	Action	Details	Timestamp	IP Address	Status
U001	Sarah Chen	Task Created	Created task "Update security protocols"	Nov 10 at 09:30 AM	192.168.1.105	Success
U002	Mike Johnson	File Uploaded	Uploaded "project-specs.pdf" (2.3 MB)	Nov 10 at 08:20 AM	192.168.1.107	Success
U003	Emma Davis	User Role Changed	Changed role for sarah.chen@university.edu to Admin	Nov 10 at 07:15 AM	192.168.1.101	Success
UNKNOWN	Unknown User	Login Failed	Failed login attempt with email admin@test.com	Nov 10 at 06:45 AM	203.45.67.89	Failed
U004	James Wilson	Task Deleted	Deleted task "Marketing Campaign Draft"	Nov 9 at 03:30 PM	192.168.1.112	Success

Showing 1 to 5 of 8 logs

< Previous

1

2

Next >



FIGMA-ADMINISTRATOR VIEW

SecureFocus

Dashboard

Chat

Files

Admin

Logout

Admin Dashboard

Secure system management and monitoring

Admin Access

Dashboard

Audit Logs

User Management

System Config

User Management

Manage user accounts, roles, and permissions

Create New User

Search users by name, email, or role...

Name	Email	Role	Status	Last Login	Actions
Sarah Chen	sarah.chen@university.edu	User	Active	Nov 10 at 09:30 AM	⋮
Mike Johnson	mike.johnson@university.edu	User	Active	Nov 10 at 0	⋮
Emma Davis	emma.davis@university.edu	Admin	Active	Nov 10 at 10	⋮
James Wilson	james.wilson@university.edu	User	Suspended	Nov 8 at 02	⋮
Lisa Anderson	lisa.anderson@university.edu	User	Active	Nov 9 at 04:30 PM	⋮

Edit Role

Reset Password

Suspend Account

Delete User

Create New User Account

×

Add a new user to the SecureFocus system

Full Name

Enter full name

University Email

user@university.edu

User Role

Standard User

Standard User

Administrator



FIGMA-ADMINISTRATOR VIEW

Admin Dashboard

Secure system management and monitoring

Admin Access

Dashboard Audit Logs User Management **System Config**

System Configuration Panel

Manage system-wide security and operational settings

Security Settings

Multi-Factor Authentication (MFA)

Require all users to enable MFA for enhanced security



Session Timeout Duration

Automatically log out inactive users after specified time

30 minutes (Default)



Audit Log Retention Period

How long to keep audit logs before archival

90 days (Default)



Project & Collaboration Settings

Default Project Visibility

Default visibility setting for newly created projects

Private (Default)



All changes saved

Reset to Defaults

Save Configuration



Threat Modeling and Key Security Findings

THREAT	RISK	MITIGATION
Spoofing	Account theft	MFA, strong passwords
Tampering	Data manipulation	Input validation
Repudiation	Denied actions	Audit logs
Information Disclosure	Data leaks	TLS, encryption
DoS	Service crash	Rate limiting
Privilege Escalation	Service crash	Code review, validation

● ● ● Threat Modeling and Key Security Findings

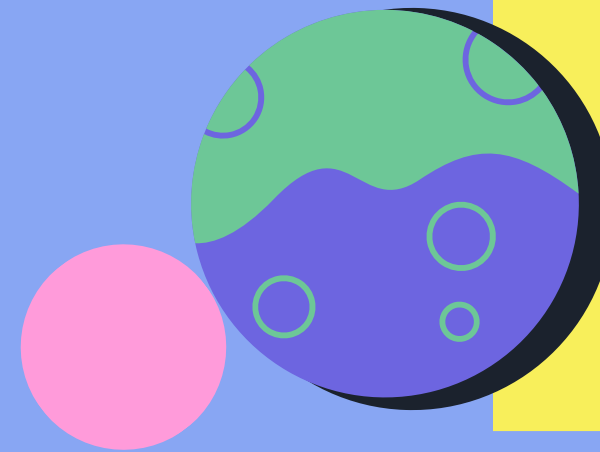
Why not others? Threat Modeling (STRIDE)

- Identified threats across login, tasks, files, and chat.
- Analyzed impacts + mapped mitigations using STRIDE.

Key

Prioritized Focus

- Authentication
- Data integrity & confidentiality
- Availability
- Accountability



Key Security Findings

Strong defenses for Spoofing & Privilege Escalation (MFA, RBAC).

Data protection via input validation + encryption.

Availability improved with rate limiting & traffic monitoring.

Accountability strengthened through audit logs.



FUTURE WORK

- Cloud deployment on platforms like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).
- Continuous Integration and Continuous Deployment (CI/CD) pipelines with automated testing.
- Conduct external penetration testing and a full security audit to validate protection measures.
- Add custom user roles and support for real-time collaboration features.
- Develop a mobile application for both Android and iOS platforms.



